AMRITA VISHWA VIDYAPEETHAM
SCHOOL OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Explainable Hybrid Bimodal Deepfake Detection for Image and Video Analysis

23CSE399 – Project Phase 1 (Panel Review 2)

Team Members		Guide	Project Type	Marks
CB.SC.U4CSE23205	Alana Mariya PC	**Dr. N Lalithamani** *Associate Professor,* *Dept. of CSE*	**Type 1** Research	.../ 5
CB.SC.U4CSE23320	Durgam Poojitha			
CB.SC.U4CSE23403	Akshay Krishna			
CB.SC.U4CSE23423	Kalyani H Karuvelil			
CB.SC.U4CSE23443	Sidharth S Nair			

1 Project Proposal & Objectives

Project Proposal: This project pioneers an Explainable AI framework designed to dismantle the black box nature of contemporary deepfake detection by bridging the gap between high-accuracy classification and human-centric interpretability. While standard deep learning models offer little more than an opaque confidence score, our multi-modal system integrates spatial and spatiotemporal feature learning to provide a granular, two fold defense against synthetic media. By synthesizing image and video analysis with real time visual explanations including heat maps that isolate discriminative facial regions and identify temporal inconsistencies the platform provides a transparent evidence trail for every detection. This shift from binary output to actionable insight not only strengthens the integrity of digital media verification but also empowers analysts to conduct targeted, high confidence inspections of suspicious content through a lens of total algorithmic transparency.

Originality Claim:

- Develop a dual-stream neural architecture that fuses spatial and spatiotemporal features to achieve high-accuracy detection across both static images and dynamic video sequences.
- Integrate Explainable AI layers to generate real-time heat maps that pinpoint the specific facial regions driving the model's classification decisions.
- Implement a temporal mapping module to isolate and visualize frame-by-frame inconsistencies, providing a clear timeline of synthetic manipulations.
- Establish a transparent diagnostic pipeline that translates opaque confidence scores into human interpretable visual justifications for improved system reliability.
- Optimize the detection framework to maintain real time processing speeds without compromising the depth or accuracy of the generated visual explanations.

2 Literature Survey & Gap Analysis

- **Existing Solutions:** Current state-of-the-art methods (e.g., Method A, Method B) focus on [Feature].
- **Limitations:** Despite significant advancements in deepfake detection, existing methods primarily rely on deep learning models that function as black boxes, offering limited interpretability beyond simple confidence scores. Most approaches focus either on spatial features in static images or temporal inconsistencies in videos, but rarely integrate both effectively into a unified framework. Additionally, current systems often lack robustness against adversarial manipulations and compression artifacts, which are common in real-world media. Another major limitation is the absence of built-in explainability mechanisms, making it difficult for users to understand or trust the model's decisions. Furthermore, many high-accuracy models suffer from high computational complexity, limiting their ability to operate in real-time while maintaining detailed analytical outputs such as visual explanations or heatmaps.

- **Research Gap:** From the literature survey, it is evident that there is no comprehensive end-to-end deepfake detection system that simultaneously addresses accuracy, robustness, and interpretability. Specifically, there is a lack of efficient frameworks that integrate spatial and spatiotemporal feature learning for both images and videos within a single architecture. Existing solutions do not adequately incorporate adversarial robustness or preprocessing techniques to handle compression-related distortions. Moreover, the absence of real-time Explainable AI (XAI) components that provide visual justifications, such as heatmaps and temporal inconsistency mapping highlights a critical gap in making these systems transparent and trustworthy. This gap underscores the need for a unified, explainable, and multimodal deepfake detection framework capable of delivering reliable, interpretable, and real-time analysis for both static and dynamic media.

3 Execution Plan

3.1 Identification of Subdomains

To address the problem of deepfake detection in images and videos, the solution is divided into the following key subdomains:

- **Computer Vision (Spatial Feature Learning):** Learning visual patterns, textures, and facial inconsistencies using XceptionNet and EfficientNet.
- **Temporal Sequence Modeling:** Understanding frame-to-frame variations in videos using LSTM.
- **Data Preprocessing:** Handling input variations using compression-based preprocessing.
- **Robust Machine Learning:** Improving model stability using adversarial robustness training.
- **Data Augmentation:** Enhancing generalization through augmentation techniques.
- **Explainable AI (XAI):** Interpreting model decisions using Grad-CAM.
- **System Integration:** Presenting results using a user interface.

3.2 Mapping Subdomains to System Components

Subdomain	Component	Role
Computer Vision	XceptionNet, EfficientNet	Spatial feature extraction
Temporal Modeling	LSTM	Sequence learning
Preprocessing	Compression-based preprocessing	Input robustness
Robust ML	Adversarial training	Handle perturbations
Data Augmentation	Augmentation module	Improve generalization
Explainability	Grad-CAM	Visual interpretation
System Design	UI	Display results

Table 1: Mapping of subdomains to system components

3.3 Execution Workflow

1. **Data Preparation:** Use image and video and datasets, perform frame extraction, and organize labels.
2. **Preprocessing:** Apply JPEG (images) and H.264 (videos) compression-based preprocessing.
3. **Training Enhancement:** Apply data augmentation and adversarial robustness training.
4. **Model Development:** Train XceptionNet for images and EfficientNet + LSTM for videos.
5. **Model Integration:** Combine outputs to generate predictions.
6. **Explainability:** Generate heatmaps using Grad-CAM.
7. **Evaluation:** Evaluate using accuracy and prediction consistency.
8. **User Interface:** Display prediction, confidence, and heatmap.

3.4 Learning Plan

- **Phase 1:** Fundamentals – CNNs, RNNs, LSTM, and image processing.
- **Phase 2:** Model understanding – XceptionNet, EfficientNet, and video processing.

- **Phase 3:** Advanced techniques – Adversarial training and Grad-CAM.
- **Phase 4:** Implementation – Model training and pipeline integration.

3.5 Final Outcome

The execution plan leads to a system that:

- Detects deepfakes in images and videos
- Handles data variations and perturbations
- Provides interpretable predictions using Grad-CAM
- Supports systematic evaluation

4 Proposed Methodology (Technical Knowledge)

Core Concept: We propose a robust Deepfake detection system that integrates advanced deep learning models with adversarial robustness, and explainable AI to classify facial images and videos into either real or fake. Based on the type of input, either image-based or video-based, the designed system uses a dual-model architecture: the XceptionNet model for the extraction of spatial features from facial images, while EfficientNet combined with LSTM models captures the temporal inconsistencies in video sequences. To enhance model reliability against real-world distortions and adversarial attacks, compression-based preprocessing and adversarial robustness training are incorporated. Furthermore, the system applies a Grad-CAM-based XAI approach, which explains its predictions visually, thus increasing transparency and trust in the model's predictions.

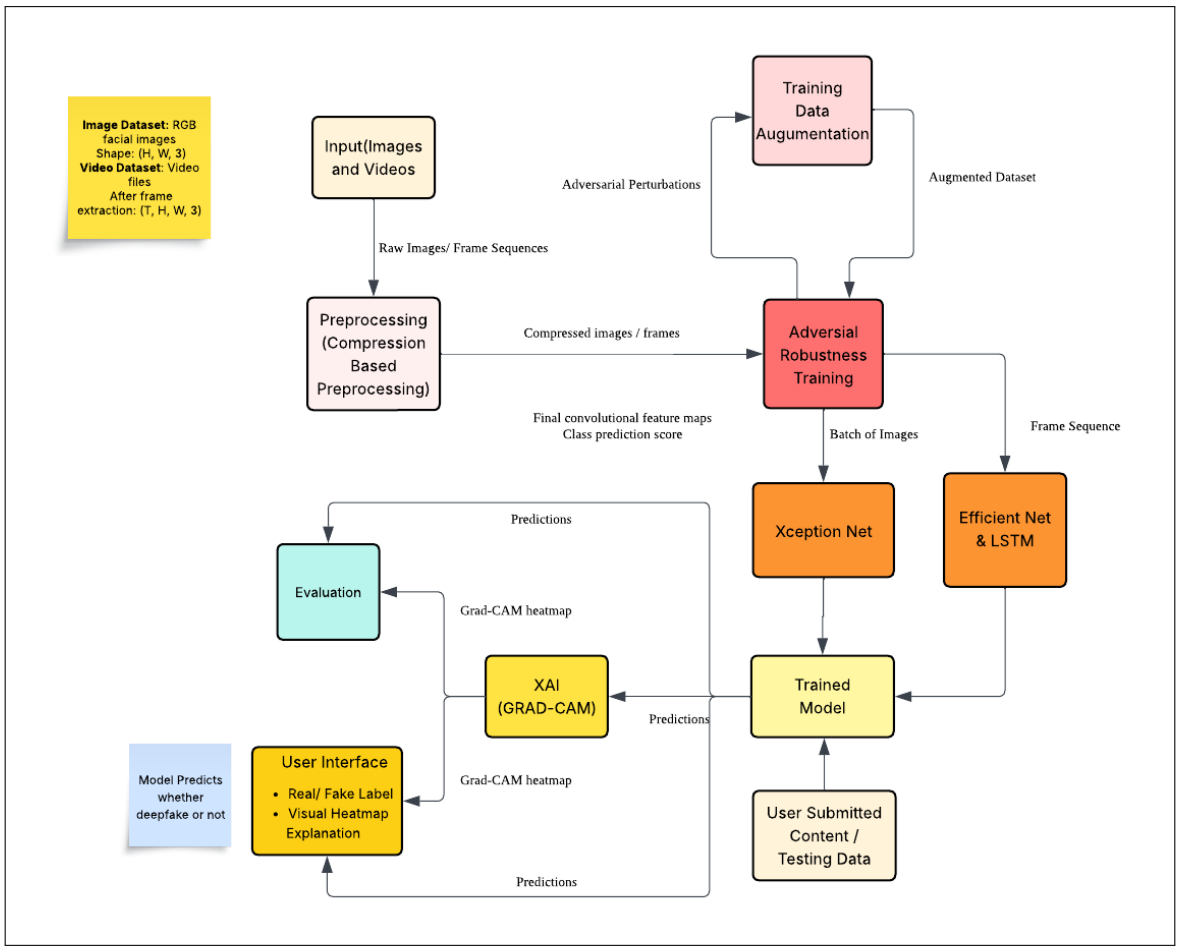


Figure 1: Proposed System Architecture

Mathematical/Logical Basis: The system accepts RGB facial images and video files as input. For video inputs, frames are first extracted to form temporal frame sequences. All inputs undergo preprocessing, which includes compression stress simulation to mimic real-world data degradation. The preprocessed data is then augmented using adversarial variations to create a robust training dataset.

During training, adversarial samples continue to be generated to improve the models resistance to manipulated inputs. Image data is passed to XceptionNet, which extracts high-level spatial features and generates class prediction scores. Video frame sequences are processed by EfficientNet for spatial feature extraction, followed by an LSTM network that models temporal dependencies across frames to detect motion-based anomalies.

Grad-CAM will be used to generate heatmaps that illustrate which areas most influence the model's decision. The trained model and the accuracy of the explanations are evaluated using quantitative performance metrics, and the final output is presented through a user interface displaying the real/fake classification label along with visual heatmap explanations.

5 Summary of Work and Future Plan

5.1 Work Completed

- Initially proposed a forensic-based deepfake detection system; later refined to a research-based approach due to data and feasibility constraints.
- Conducted literature review on CNNs, LSTMs, Vision Transformers, and XAI methods.
- Identified research gaps such as lack of robustness, explainability, and integrated frameworks.
- Learned key concepts including XceptionNet, EfficientNet, compression-based preprocessing, adversarial training, and Grad-CAM.
- Proposed system architecture integrating image and video detection with explainability.

5.2 Research Findings

- Existing methods focus mainly on accuracy and lack robustness and interpretability.
- Need for a unified deepfake detection pipeline.

5.3 Future Plan

- **Second Phase:** Partial implementation including preprocessing, XceptionNet, and EfficientNet + LSTM models.
- **Third Phase:** Full system integration with adversarial training, Grad-CAM, evaluation, and UI development. Preparation of final report, research paper and publication of research paper

References

[1] M. S. Momin, A. Sufian, D. Barman, M. Leo, C. Distante, and N. Damer, "Explainable deepfake detection across different modalities: An overview of methods and challenges," *Image and Vision Computing*, pp. 105 738–105 738, 09 2025.

[2] A. Almestekawy, H. H. Zayed, and A. Taha, "Deepfake detection: Enhancing performance with spatiotemporal texture and deep learning feature fusion," *Egyptian Informatics Journal*, vol. 27, p. 100535, 09 2024.

[3] S. Mathews, S. Trivedi, A. House, S. Povolny, and C. Fralick, "An explainable deepfake detection framework on a novel unconstrained dataset," *Complex & Intelligent Systems*, vol. 9, 01 2023.

[4] N. Mansoor and A. I. Iliev, "Explainable ai for deepfake detection," *Applied Sciences*, vol. 15, pp. 725–725, 01 2025. [Online]. Available: https://www.mdpi.com/2076-3417/15/2/725

[5] M. M. Alwateer, "Explainable deep fake framework for images creation and classification," *Journal of Computer and Communications*, vol. 12, pp. 86–101, 2024.

[6] L. H. Singh, P. Charanarur, and N. K. Chaudhary, "Advancements in detecting deepfakes: Ai algorithms and future prospects - a review," *Discover Internet of Things*, vol. 5, 05 2025.

[7] M. Masood, M. Nawaz, K. M. Malik, A. Javed, A. Irtaza, and H. Malik, "Deepfakes generation and detection: State-of-the-art, open challenges, countermeasures, and way forward," *Applied Intelligence*, vol. 53, 06 2022. [Online]. Available: https://link.springer.com/article/10.1007/s10489-022-03766-z

[8] R. U. Maheshwari and B. Paulchamy, "Securing online integrity: a hybrid approach to deepfake detection and removal using explainable ai and adversarial robustness training," *Automatika*, vol. 65, pp. 1517–1532, 09 2024.

[9] S. Venkateswarulu and A. Srinagesh, "Deepexplain: Enhancing deepfake detection through transparent and explainable ai model," *Informatica*, vol. 48, 05 2024.

[10] B. Gowrisankar and V. L. Thing, "An adversarial attack approach for explainable ai evaluation on deepfake detection models," *Computers & security*, vol. 139, pp. 103 684–103 684, 04 2024.

[11] S. Ali, T. Abuhmed, S. El-Sappagh, K. Muhammad, J. M. Alonso-Moral, R. Confalonieri, R. Guidotti, J. D. Ser, N. Díaz-Rodríguez, and F. Herrera, "Explainable artificial intelligence (xai): What we know and what is left to attain trustworthy artificial intelligence," *Information Fusion*, vol. 99, p. 101805, 04 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1566253523001148

[12] F. Khalid, A. Javed, K. M. Malik, and A. Irtaza, "Explanet: A descriptive framework for detecting deepfakes with interpretable prototypes," *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 6, pp. 486–497, 10 2024.

[13] R. Rafique, R. Gantassi, R. Amin, J. Frnda, A. Mustapha, and A. H. Alshehri, "Deep fake detection and classification using error-level analysis and deep learning," *Scientific Reports*, vol. 13, p. 7422, 05 2023. [Online]. Available: https://www.nature.com/articles/s41598-023-34629-3

[14] S. A. Khan and D.-T. Dang-Nguyen, "Deepfake detection: Analysing model generalisation across architectures, datasets and pre-training paradigms," *IEEE Access*, vol. 12, pp. 1880–1908, 01 2024.

[15] G. Gupta, K. Raja, M. Gupta, T. Jan, S. T. Whiteside, and M. Prasad, "A comprehensive review of deepfake detection using advanced machine learning and fusion methods," *Electronics*, vol. 13, pp. 95–95, 12 2023.

[16] I. Balafrej and M. Dahmane, "Enhancing practicality and efficiency of deepfake detection," *Scientific Reports*, vol. 14, 12 2024.

[17] A. H. Soudy, O. Sayed, H. Tag-Elser, R. Ragab, S. Mohsen, T. Mostafa, A. A. Abohany, and S. O. Slim, "Deepfake detection using convolutional vision transformers and convolutional neural networks," *Neural Computing and Applications*, vol. 36, 08 2024.

[18] T. Qiao, S. Xie, Y. Chen, F. Retraint, R. Shi, and X. Luo, "Deepfake detection fighting against noisy label attack," *IEEE Transactions on Multimedia*, vol. 26, pp. 9047–9059, 01 2024.

[19] Y. Xu, P. Terhörst, M. Pedersen, and K. Raja, "Analyzing fairness in deepfake detection with massively annotated databases," *IEEE Transactions on Technology and Society*, vol. 5, pp. 1–1, 2024.

[20] K. N. Ramadhani, R. Munir, and N. P. Utama, "Improving video vision transformer for deepfake video detection using facial landmark, depthwise separable convolution and self attention," *IEEE Access*, vol. 12, pp. 8932–8939, 01 2024.

[21] L. A. Motalib, O. Mustapha, and H. Mustapha, "Compression-aware hybrid framework for deep fake detection in low-quality video," *IEEE Access*, vol. 13, pp. 131 980–131 997, 2025.

[22] D. Xiong, Z. Wen, C. Zhang, D. Ren, and W. Li, "Bmnet: Enhancing deepfake detection through bilstm and multi-head self-attention mechanism," *IEEE Access*, vol. PP, p. 1, 01 2025.

[23] X. Liu, Y. Yu, X. Li, and Y. Zhao, "Mcl: Multimodal contrastive learning for deepfake detection," *IEEE transactions on circuits and systems for video technology*, vol. 34, pp. 1–1, 01 2023.

[24] X. Jin, Y. Kou, Y. Xie, Y. Zhao, M. Laiha, Q. Jiang, and W. Zhou, "Learning local texture and global frequency clues for face forgery detection," *Biomimetics*, vol. 10, pp. 480–480, 07 2025.

[25] L. Cirillo, A. Gervasio, and I. Amerini, "Explainability-driven adversarial robustness assessment for generalized deepfake detectors," *EURASIP Journal on Information Security*, vol. 2025, 08 2025.

[26] P. Sharma, M. Kumar, and H. K. Sharma, "A robust ensemble model for deepfake detection of gan-generated images on social media," *Discover Computing*, vol. 28, 04 2025.

[27] A. Abul Hasanaath, H. Luqman, R. Katib, and S. Anwar, "Fsbi: Deepfake detection with frequency enhanced self-blended images," *Image and Vision Computing*, vol. 154, p. 105418, 01 2025.

[28] R. Sunil, P. Mer, A. Diwan, R. Mahadeva, and A. Sharma, "Exploring autonomous methods for deepfake detection: A detailed survey on techniques and evaluation," *Heliyon*, vol. 11, no. 3, Feb 2025.